



ME 300:
Engineering Thermodynamics

Introduction to the Second Law

Energy conservation

$$\left. \begin{aligned} \Delta E &= Q - \bar{W} \\ \frac{dE}{dt} &= \dot{Q} - \dot{\bar{W}} \end{aligned} \right\} \begin{aligned} &- \text{energy is conserved} \\ &- \text{way energy is exchanged} \begin{aligned} &\rightarrow \text{heat} \\ &\rightarrow \text{work} \end{aligned} \end{aligned}$$

$\Rightarrow$ 2nd law $\rightarrow$ limits

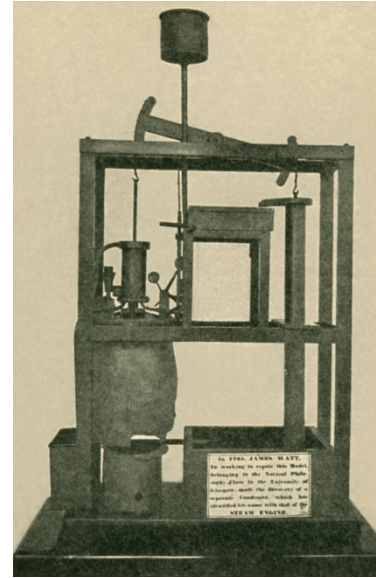


How does the first law prescribe the direction of the process?

→ No

Newcomen (1712) & Watt (1765)

- Technology was developed much earlier than the theory was available to explain it
- Newcomen and Watt developed and further improved upon the steam engine



Sadi Carnot (1824)

- Carnot was a military engineer who, being frustrated with the state of steam engine technology, wrote a paper on the fundamental principles of the Second Law

